Risk Assessment and Mitigation Plan

Project 2: Banking Transaction System

Milestone 1 – Risk Assessment and Mitigation Plan

1. Purpose

This document identifies, analyzes, and plans responses to the risks that could affect the delivery of the Banking Transaction System: a Java application handling accounts, deposits, withdrawals, and transaction history, together with manual and automated testing and Jira-based test management.

2. Scope

The assessment covers the full project lifecycle: development (OOP, collections, exception handling, file/database handling), testing (manual plans, static testing, Selenium WebDriver, Postman), and reporting in Jira.

3. Risk Assessment Method

Each risk is rated on Probability (P) and Impact (I) using a 1–3 scale (1 = Low, 2 = Medium, 3 = High). Risk Score = P × I.

Score	Level	Response
1–2	Low	Monitor
3–4	Medium	Mitigate and track
6–9	High	Immediate mitigation plan and owner

4. Risk Register

ID	Category	Risk	P	I	Score	Level
R1	Technical	Incorrect balance calculation in deposits/ withdrawals	2	3	6	High
R2	Technical	Unhandled invalid transactions (negative amounts, overdraft, non-existent account)	2	3	6	High
R3	Data	Data loss or corruption in file/database storage	2	3	6	High
R4	Data	Database connection or schema issues (MySQL/ SQLite)	2	2	4	Medium
R5	Security	Weak validation allowing unauthorized or malformed operations	2	3	6	High
R6	Testing	Incomplete test coverage of edge cases	3	2	6	High

R7	Testing	Unstable Selenium tests (timing, locator changes)	3	2	6	High
R9	Schedule	Tasks underestimated; deadlines missed	3	2	6	High
R10	Team	Unclear roles or member unavailability	2	2	4	Medium
R11	Team	Merge conflicts and inconsistent code between members	2	2	4	Medium
R12	Tools	Jira misuse: bugs not logged or tracked properly	2	1	2	Low
R13	Scope	Scope creep (extra features beyond requirements)	2	2	4	Medium
R14	Environment	Tool or version incompatibility (Java, drivers, browsers)	2	2	4	Medium

5. Mitigation Plan

ID	Mitigation (prevention)	Contingency (if it happens)	Owner
R1	Unit tests for every transaction type; code review of calculation logic	Fix, retest, and log defect in Jira	Developer
R2	Custom exceptions (e.g., InsufficientFundsException, InvalidAmountException) and input validation	Add missing validation and regression tests	Developer
R3	Use transactions/commits; regular backups of data files	Restore from backup; replay from transaction history	Developer
R4	Set up test DB early; keep a schema script in the repository	Switch to SQLite fallback	Developer
R5	Validate all inputs; restrict operations by account state	Patch and run security-focused test cases	Developer / Tester
R6	Boundary value analysis and equivalence partitioning; review test cases	Add tests for uncovered areas before milestone end	Tester
R7	Explicit waits, stable locators, Page Object Model	Rerun, isolate flaky tests, fix locators	Automation Tester
R8	Keep Postman collections versioned; update with each API change	Update collection and rerun	Tester
R9	Break work into small tasks; weekly progress check; buffer time	Reassign tasks; reduce lower-priority items	Project Manager
R10	Written role assignment; backup owner per task	Redistribute tasks at team meeting	Project Manager
R11	Git branches and pull requests; small frequent commits	Resolve conflicts together; agree on code conventions	All members

R12	Define Jira workflow and bug template; log defects immediately	Update tickets before reporting	Tester

6. Monitoring and Review

- Review the risk register weekly in the team meeting.
- Update probability, impact, and status as the project progresses.
- Log materialized risks and their resolution in Jira.
- Add newly discovered risks with ID, owner, and mitigation.

7. Risk-Related KPIs

KPI	Target
High-level risks with an assigned owner and mitigation	100%
Risk register reviewed on schedule	Weekly
Defects traced to a registered risk	Tracked and decreasing
Mitigation actions completed on time	≥ 90%

8. Assumptions

- Team roles follow the Task Assignment and Roles document.
- Owners listed above are placeholders; replace them with team member names.